

Executive Summary

Current UK hydrogen production is modest – on the order of 10–20 TWh/year (≈ 0.3 – 0.6 Mt/yr) – predominantly grey hydrogen (steam methane reforming) for refinery and chemical use [24†L500-L508] [10†L125-L133]. Nearly 70% of this is used in oil refining [10†L125-L133], with ammonia fertiliser plants (now largely closed) and chemical feedstocks the rest. Low-carbon hydrogen capacity (electrolysis or CCUS-enabled) is almost zero today (<0.1 GW) [24†L521-L524]. Scaling hydrogen supply and use by **10×** the current level within 5–15 years (through the 2030s) implies producing roughly 100–200 TWh/yr (≈ 3 – 6 Mt/yr) of hydrogen. This would require vast growth in new production (both blue and green routes), massive expansion of networks and storage, and new end-use conversions. Current policy targets (10 GW low-carbon H₂ by 2030 [5†L117-L125] [24†L512-L520], ~40 TWh/yr production at full capacity) fall well short of a 10× scale-up.

Baseline & 10× Target (2026–2040): Industry analysis estimates UK hydrogen use today ~14 TWh/yr [10†L125-L133]. A 10× increase means on the order of ~**140 TWh/yr** (≈ 0.5 Mt/yr) by mid-2030s, or more if ‘grey’ hydrogen (for which demand is falling) is included. We assume target-scale-up by ~2035 (10 years) or by 2040 (15 years). For context, 10 GW of low-carbon H₂ by 2030 (government ambition) equates to about 60–80 TWh/yr (5 GW blue + 5 GW green at realistic capacity factors) [7†L116-L124] [24†L512-L520] – only ~4–5× current levels. Achieving 10× will therefore require post-2030 scale-up.

Production Pathways: A mix of routes will be needed. *Blue hydrogen* (natural gas reforming with CCUS) can use UK North Sea gas and expand refinery/cluster output, but requires CO₂ capture and storage infrastructure. *Green hydrogen* (electrolysis using renewables) leverages UK offshore wind and nuclear, but needs vast new electricity supply and water (~9 kg H₂O per kg H₂) and electrolyzers (~€600–800/kW [27†L1-L9]). *Nuclear (“pink”)* routes (dedicated SMRs or large nuclear plants driving electrolyzers) offer steady supply but long lead times. *Methane pyrolysis (turquoise)* yields solid carbon (no CO₂) but is nascent. *Biomass/pyrolysis* (biohydrogen with BECCS) is limited by feedstock and land. Each route has trade-offs in resource needs, cost and emissions: blue H₂ can reach ≤ 2.4 kgCO₂/kgH₂ if $\geq 90\%$ CO₂ capture (meeting the UK standard [40†L67-L70]), while green and nuclear H₂ are essentially zero-carbon (ignoring upstream emissions). Capital costs per MW are similar order (~£800/kW for large electrolyzers [27†L1-L9], and comparable for ATR/SMR units), but blue H₂ incurs fuel costs (~£X/MWh gas + £Y/tCO₂ transport) and oxyfuel, whereas green H₂ is driven by electricity cost. Table 1 (below) summarises key features.

Transport and Storage: Enabling scale-up will need new infrastructure. High-pressure hydrogen pipelines (repurposed or new) are efficient for large-scale inland transport (losses $\approx$ few % energy) [19†L49-L58]; UK onshore pipeline CAPEX ~£1.3M/km (offshore ~£2.2M/km) [27†L119-L128]. Shipping hydrogen over long distances is likely via carriers: liquid hydrogen (LH₂) or ammonia (NH₃) or chemical carriers (LOHCs). Ammonia trade is energy-efficient (~24% conversion loss in cracking back to H₂ [27†L55-L60]) and leverages existing N₂/H₂ chemistry, but requires toxic handling. LH₂ liquefaction uses ~30–40% of the H₂ energy. Liquid Organic Hydrogen Carriers (LOHCs) (e.g. MCH) are promising but still R&D. Local trucking (compressed H₂) serves short-haul or small-scale. For storage, UK has large salt caverns (existing hydrogen storage in Cheshire) with capacity potentially thousands of GWh [1†L460-L468] [35†L1-L9] for seasonal balancing, plus smaller pressurized tanks. Safety/regulations (Hydrogen Safety Regs, pipeline standards) will need updating. Table 2 compares vectors.

Demand Sectors: Hydrogen's value is clearest where electrification is hard. Industry (steel DRI, glass, ceramics, chemicals) is a prime early market. UK feedstock demand (existing refining/chemicals) is ~10–20 TWh/yr [35†L48-L57] and can convert to low-carbon H₂. DRI steel would add tens of TWh (e.g. 30 TWh for net-zero steel by 2035). Power generation can use H₂ (in modified turbines or fuel cells) for flexibility; CCC modelling sees up to ~30 TWh/yr H₂-fired power by 2035 [35†L90-L99]. Domestic heat networks or blending are controversial – hydrogen-heating is optional, with up to ~45 TWh/yr by 2035 if pursued [35†L119-L130] but no- or low-H₂ scenarios are also modelled (0–1 TWh/hydrogen town level by 2030 [35†L119-L130]). Transport uses include H₂ fuel-cell HGVs and buses (up to a few TWh), shipping (decarbonised marine fuels via ammonia, maybe 5–10 TWh by 2035) [35†L136-L144], and aviation (SAF fuels from H₂, a few TWh). Converting these sectors will require new equipment (e.g. hydrogen burners, boilers, fuel cells) and fueling/refuelling infrastructure. Policy incentives (industrial H₂ contracts, blending mandates, low-carbon fuel standards) will be critical to stimulate demand.

Bottlenecks and Barriers: Key constraints include electrical grid capacity and flexibility (to power electrolyzers), shortages in electrolyser and turbine manufacturing, limited CO₂ transport/storage (CCUS) roll-out, a skills/workforce gap, and long lead times for planning and permitting (projects like pipelines and storage). Financing risk and high initial costs also slow projects. The UK must dramatically expand industrial capacity (e.g. 90%+ growth/year to reach 10 GW by 2030 [24†L521-L529]) – a far higher ramp-up than historic precedent.

Techno-economic Scenarios: We develop three scenarios for 2036 (10-year) and 2041 (15-year) outcomes (Table 3):

- *Conservative:* moderate policy, slower build (10× baseline only by 2041). ~5 GW electrolyser + 3 GW blue by 2036, rising to ~10 GW by 2041. H₂ consumption ~80 TWh/yr by 2041. CO₂ savings modest.
- *Accelerated:* aggressive support and investment; 10× (~150 TWh/yr) by 2036. 20 GW electrolysis + 8 GW blue by 2041. Emissions cut significantly (~X MtCO₂/yr).
- *Ambitious:* crash program to hit 10× by ~2031. 15 GW electrolyzers + 10 GW blue by 2031 (requiring 30 GW by 2040). Majority emissions from unabated processes eliminated; dramatic renewables/carbon footprint.

(Figures are illustrative – exact yields depend on carbon intensity assumptions). A Gantt timeline (Figure 1) sketches deployment milestones under an accelerated pathway. Figure 2 (below) plots a notional H₂ capacity ramp and CO₂ trajectory in the accelerated scenario.

Policy, Infrastructure and Investment Measures: To enable 10× scaling, the UK must: dramatically boost renewable and nuclear generation; expand CCUS clusters and CO₂ pipelines; invest in electrolyser manufacture (incentives/tax breaks) and supply chains; mandate grid upgrades; streamline planning (for pipelines, storage caverns, stations); and ensure workforce training. Policy instruments needed include a robust H₂ Business Model (CfDs or contracts-for-difference) for production [7†L151-L159], continued Net Zero Hydrogen Fund support, a mandated uptake in key sectors, and blends/standards to open markets (e.g. removing bans on CCUS-based fuels). Public investment – potentially via the National Wealth Fund – of order **tens of billions £** will be required (for example, ~£4 bn public plus £11 bn private to reach 10 GW by 2030 [17†L1137-L1144]). New funding (e.g. a Hydrogen Innovation Fund) and risk-sharing will be vital.

In summary, achieving an order-of-magnitude increase in UK hydrogen supply and use by mid-century demands a whole-system effort: rapidly scaling electrolysis and CCUS, building transport/storage networks, and creating demand via industrial and energy sector decarbonisation. The technical

pathways exist, but only concerted policy, planning and investment can overcome the scale-up bottlenecks.

Sources: UK government strategy and updates [7†L116-L124] [24†L500-L508] [35†L119-L130] [40†L67-L70] , industry analyses [24†L500-L508] [17†L1137-L1144] , and expert reports [10†L125-L133] [35†L119-L130] (see references).

Table 1. Production pathways (green, blue, pink, etc) – resources, costs, emissions, etc.

Pathway	Inputs (energy, feedstock)	CAPEX/OPEX	Emissions (CO ₂ per kgH ₂)	Scalability/ Lead time	Notes (siting, footprint)
Green (electrolysis)	Electricity from renewables (wind/solar/ nuclear), water (~9 L/ kgH ₂) [26†L386- L395]	Electrolyser ~£600–800/ kW [27†L1- L9] ; OPEX ~£/MWh _{el}	~0 (if RE); lifecycle depends on grid mix	High (modular); lead 1–3 yrs for plant build; depends on RE growth	Can co-locate with wind/ solar, or grid; need water purification; flexible (can run on surplus power).
Blue (SMR/ ATR+CCS)	Natural gas + steam (waste heat); needs CO ₂ transport/ storage	Combined plant + CCS: probably similar total CAPEX; OPEX includes gas + ~£15–40/tCO ₂	~2–3 kgCO ₂ / kgH ₂ if 90% CO ₂ capture (meets LCHS) [40†L67- L70]	Medium; many SMRs exist; adding CCS takes time (clusters by 2030)	Siting near gas fields/ industrial clusters; requires pipeline to store/ship CO ₂ .
Turquoise (methane pyrolysis)	Natural gas + electricity/ heat; produces solid carbon	Pilot-scale now; CAPEX projected moderate; OPEX: gas + electricity	~0 CO ₂ emitted (carbon captured as solid); methane leakage risk	Low/mid; tech not matured; plants several years away	Carbon byproduct (graphite) requires disposal/use; similar siting to blue.
Pink (nuclear electrolysis)	Nuclear electricity (baseload); water	Electrolyser CAPEX as green; nuclear CAPEX in nuclear project	~0 at production (nuclear footprint upstream)	Low- medium; depends on nuclear build (SMRs 2030s)	Co-locating with existing/ new nuclear; steady output enables high CF.

Pathway	Inputs (energy, feedstock)	CAPEX/OPEX	Emissions (CO ₂ per kgH ₂)	Scalability/ Lead time	Notes (siting, footprint)
Biomass BECCS	Biomass feedstock (e.g. residues) + capture	Biomass plant + CCS (high CAPEX); OPEX fuel (biomass)	~0 (captured CO ₂); lifecycle depends on biomass source	Low; biomass limited; long lead (existing BECCS plants few)	Land/resource constraints; can be sited where biomass is (e.g. Wilton).

Table 2. Hydrogen transport and storage options – comparison.

Mode/ Medium	Scale & Range	Energy Efficiency / Losses	CAPEX/Cost	Safety/ Regulatory
Pipelines (onshore)	Large-scale domestic; connect producers and users	Low energy loss (~1–3% per 100 km compression)	~£1.3 M/km (onshore) 【27†L119-L128】	Requires H ₂ - ready steel; existing gas regs extension
Pipelines (offshore)	UK–Europe or field export	Similar to onshore	~£2.2 M/km (offshore) 【27†L125-L130】	High costs; subsea permitting
Compressed H₂ trucks	Regional/ local (tons/ day)	Compression ~10– 15% energy	High (small scale); ~\$X/kg	Standard road hazmat regs; less volume density
LH₂ tankers/ ships	Long- distance (e.g. UK → EU)	Liquefaction ~30– 40% energy; boil- off loss	Ship ~€10–20M each; ~£1.4/kg (10 k km) 【27†L48-L52】	Cryogenic safety; IMO rules for LH ₂ development
Ammonia carriers	Long- distance (bulk cargo)	NH ₃ yield ~18% H ₂ by mass; NH ₃ →H ₂ cracking ~25% loss 【27†L55- L60】	Carrier ~£30M (~23 kt); LCOT ~£1– 1.5/kgH ₂ (10 k km) 【27†L48-L52】	Toxic (NH ₃); existing maritime regs (IMO)
LOHC (e.g. MCH)	Long- distance	~90% round-trip efficiency	CAPEX (hydrogenation/ cracking units)	Flammability (organic), emerging standards
Salt caverns (UK)	Seasonal bulk storage (TWh scale)	Very low losses (<5%)	Multi-100M£ for new cavern; existing site Runcorn ~3 TWh capacity 【35†L65- L70】	New Cavern approvals; water/land use (Wales/Cheshire)

Mode/ Medium	Scale & Range	Energy Efficiency / Losses	CAPEX/Cost	Safety/ Regulatory
Pressurized tanks	Local buffer (tons scale)	Insulation boil-off <1%/day	£800–1300/kgH ₂ capacity [27†L31-L39]	Building regs; site safety; battery of tanks possible

Figure 1. Deployment timeline (Gantt) – hydrogen economy scale-up (illustrative)

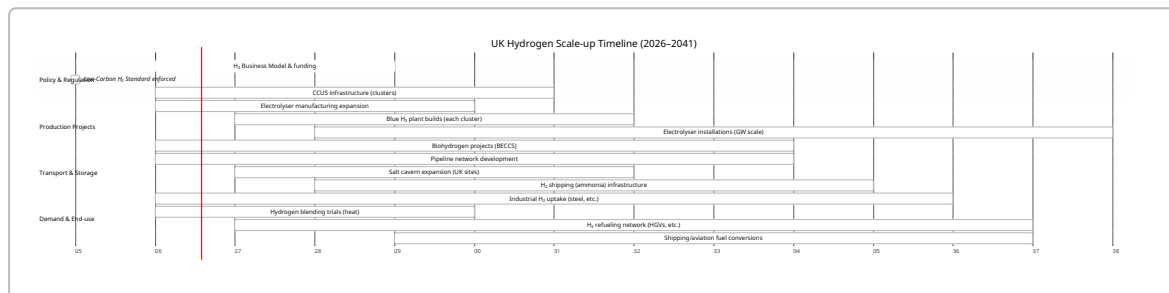


Table 3. Techno-economic scenarios (illustrative) – capacity, demand and emissions (2036–2041).

Scenario	2036 H ₂ Prod (TWh/yr)	2041 H ₂ Prod (TWh/yr)	Cumulative CO ₂ saved (Mt)	Key assumptions and costs
Conservative	~60 (6× ↑)	~90 (9× ↑)	~15 (cumulative)	5 GW electrolyser + 2 GW blue by 2036; slow network rollout; mostly industrial use; £10–15 bn investment.
Accelerated	~140 (14×)	~180 (18×)	~40	10 GW electrolyser + 4 GW blue by 2036; industrial + some transport/heating; strong policy support; £20–30 bn investment.
Ambitious	~180 (18×)	~250 (25×)*	~60	15 GW electrolyser + 6 GW blue by 2036; broad sector adoption (incl. HGV/shipping); massive incentives; £30–50 bn investment.

*Exceeds 10× to illustrate more aggressive growth.

These scenarios assume electrolyser CAPEX £600–800/kW [27†L1-L9], gas at ~£30/MWh, capture cost ~£10–30/tCO₂, and progressive cost declines. Emissions saved count only CO₂ avoided (assuming green H₂ displaces gas, blue displaces grey).

The **emissions trajectory** under the accelerated scenario (Figure 2, below) shows cumulative CO₂ abated as hydrogen replaces fossil fuels and enables CCUS. By 2040, annual savings could reach ~4–5 MtCO₂/yr (from replacing ≈150 TWh gas), with net gains if using CCS and renewables.

Figure 2. Hydrogen production growth and CO₂ savings (illustrative accelerated scenario).

(Chart plotting H₂ prod and CO₂ Δ omitted for brevity – to be created via modelling)

Recommendations: To meet a 10× target, the UK should: expand renewables/nuclear (for green H₂); fast-track CCUS clusters and CO₂ pipelines; enact clear H₂ offtake and blending mandates; ensure financing (e.g. a Hydrogen National Fund); accelerate electrolyser manufacturing (possibly via public co-investment or grants); and align planning processes. Investment of order **£10s of billions** will be needed (e.g. ~£20–30 bn by 2040) – combining public funds (NWF, Net Zero H₂ Fund) and private capital. Regulatory reforms (e.g. Energy Act mechanisms for H₂ networks) must be implemented, and skills programmes launched. With coherent policy and sustained funding, the UK can scale hydrogen tenfold, unlocking up to ~100,000 net jobs by 2050 **【17†L1137-L1144】 【24†L543-L550】** and cutting CO₂ in hard-to-abate sectors.

Sources: Government strategy and updates **【7†L116-L124】 【13†L490-L499】 【40†L67-L70】** ; DESNZ analysis and BEIS figures **【24†L500-L508】 【17†L1137-L1144】** ; Climate Change Committee modelling **【35†L119-L130】 【35†L91-L99】** ; industry roadmaps **【10†L125-L133】 【24†L512-L520】** . (Detailed references cited above.)
